



Enter the Registered Number

--	--	--	--	--	--	--	--	--

Series



EI/FOI/616

2026

**Paper-II – Industrial and
Occupational Safety, Health and
Environment
(English Version)**

Question Booklet Sl. No.

1033049



Time : 150 Minutes

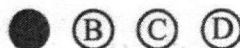
Max. Marks : 300

INSTRUCTIONS

1. The Question Paper contains **150** questions. Each question carries **2** marks. For each wrong answer **1/3rd** mark will be deducted.
2. Please check the Question Paper and ensure that it contains all the questions. If you find any defect in the Question Paper or OMR Answer Sheet, please get it replaced immediately. The OMR Answer Sheet is in Duplicate. Do not separate the copies. The markings on the first page will automatically come on the second copy also. The first copy (Original) will be retained for evaluation and the second copy (Duplicate) can be taken by the candidate.
3. Write your Registered Number on the top of the Question Paper. On the right hand corner of the first page, the Question Paper Series is printed as **(A)** or **(B)** or **(C)** or **(D)**. On the OMR Answer Sheet, Side-1 at the right top corner at Part C, the Question Paper Series are printed as **(A) (B) (C) (D)**. Darken the appropriate circle, as per your Question Paper Series with **Blue/Black Ball point pen**. Gel Pens/Pencils are not allowed.

Example to fill up the Question Paper Series on the OMR Answer Sheet.

If your Question Paper Series is A, please fill as shown below :



If you have not marked the Question Paper Series on the OMR Answer Sheet, Side-1 at Part C with Blue/Black Ball point pen, or marked in a way that it leads to discrepancy in determining the exact Question Paper Series, then your Answer Sheet will be invalidated without any further notice. No correspondence will be entertained in the matter.

4. Please get the signature of the Invigilator affixed in the space provided at the top of the OMR Answer Sheet. The OMR Answer Sheet without the signature of the Invigilator will be invalidated.



5. Each question is followed by 4 answer choices. Of these, you have to carefully select one correct answer and mark it on the OMR Answer Sheet by completely darkening the appropriate circle for the question. If more than one circle is darkened, the answer will not be valued at all. Use Blue or Black Ball point pen to make thick mark to fill the circle completely. Make **no** other stains or marks. Using **Whitener/Blade/Eraser/Chalk Powder** anywhere on the OMR Answer Sheet or any kind of tampering to change the answers on OMR Answer Sheet will lead to **invalidation** of the entire OMR Answer Sheet.

e.g. : If the answer for Question No. 1 is Answer choice (2), it should be marked as follows :

1	①	●	③	④
---	---	---	---	---

6. Do **not** fold, tear, wrinkle, tie, staple, do any rough work or make any stray marks on the OMR Answer Sheet. Otherwise your OMR Answer Sheet will be **invalidated**.
7. Do **not** mark answer choices on the Question Paper. Violation of this will be viewed seriously.
8. Rough work may be done on the last page of the Question Paper only. It should not be done anywhere else.
9. Any type of Electronic Devices such as Calculators, Mathematical/Log Tables, Mobile Phones, Bluetooth instruments etc., should not be brought into the Examination Hall. If found in possession of such devices, it will be considered as cheating and strict action taken.
10. As per G.O. Ms. No. 385, GA (Ser.A) Dept., dt: 18/10/2016, if any candidate attempts to use any means to cheat/copy etc., he/she is liable for prosecution (Police Case) and debarment.
11. Candidates should stay in the Examination Hall till expiry of Full Time i.e. 2½ Hours. No candidate shall be allowed to leave the Hall in the middle of the examination under any circumstances.
12. Before leaving the examination hall, the candidate should hand over the OMR Answer Sheet top sheet (Original) to the Invigilator and carry the bottom sheet (Duplicate) for his/her record. Carrying away Original OMR Answer Sheet will be a criminal offence.



1. In safety policy formulation, the most critical management responsibility is
 - (1) Allocation of PPE
 - (2) Visible leadership commitment
 - (3) Accident reporting
 - (4) Statutory compliance
2. The concept of inherent safety emphasizes
 - (1) Reducing consequences
 - (2) Eliminating hazards at source
 - (3) Using protective equipment
 - (4) Emergency preparedness
3. Which principle justifies stopping an activity even when scientific certainty is lacking ?
 - (1) Polluter pays principle
 - (2) Precautionary principle
 - (3) Cost-benefit principle
 - (4) Sustainability principle
4. A lagging safety indicator is best represented by
 - (1) Unsafe act observation rate
 - (2) Safety training hours
 - (3) Lost time injury rate
 - (4) Near-miss reports
5. Occupational exposure limits are primarily based on
 - (1) Economic feasibility
 - (2) Toxicological and epidemiological data
 - (3) Industrial productivity
 - (4) Worker perception
6. Safety culture maturity is highest when safety is viewed as
 - (1) A rule-based requirement
 - (2) A management responsibility
 - (3) A core organizational value
 - (4) A regulatory obligation
7. Which hazard category includes repetitive strain injuries ?
 - (1) Chemical
 - (2) Biological
 - (3) Ergonomic
 - (4) Physical
8. Environmental auditing differs from EIA because auditing is
 - (1) Predictive
 - (2) Post-implementation
 - (3) Optional
 - (4) Project-specific





9. The Factories Act places primary responsibility for worker safety on

- (1) Inspector of factories
- (2) Safety officer
- (3) Occupier
- (4) Workers

10. Chronic exposure to benzene primarily affects

- (1) Nervous system
- (2) Respiratory tract
- (3) Hematopoietic system
- (4) Digestive system

11. Risk is mathematically expressed as

- (1) Hazard $\times$ exposure
- (2) Probability $\times$ consequence
- (3) Exposure $\div$ severity
- (4) Consequence $\div$ frequency

12. A Gaussian plume model becomes unreliable during

- (1) Neutral conditions
- (2) Unstable atmosphere
- (3) Complex terrain
- (4) Steady wind

13. Which factor most influences toxic dispersion distance?

- (1) Molecular weight
- (2) Wind speed
- (3) Release temperature
- (4) Ambient humidity

14. FN curves are primarily used to represent

- (1) Individual risk
- (2) Societal risk
- (3) Environmental damage
- (4) Financial risk

15. Fault Tree Analysis fails to capture

- (1) Logical failure paths
- (2) Sequential accident evolution
- (3) Root causes
- (4) Boolean logic

16. Event Tree Analysis is most suitable for

- (1) Identifying failure causes
- (2) Modeling accident outcomes
- (3) Equipment reliability
- (4) Maintenance planning



17. Which uncertainty is irreducible in QRA ?

- (1) Model uncertainty
- (2) Parameter uncertainty
- (3) Aleatory uncertainty
- (4) Data uncertainty

18. BLEVE severity increases primarily with

- (1) Vessel diameter
- (2) Liquid superheat
- (3) Wind velocity
- (4) Ground roughness

19. Source term modeling defines



- (1) Toxicity limits
- (2) Release rate and duration
- (3) Exposure criteria
- (4) Emergency zone

20. A risk matrix limitation is

- (1) Excessive conservatism
- (2) Subjective categorization
- (3) Numerical complexity
- (4) Data requirement



21. Individual risk contours represent

- (1) Probability of fatality per year
- (2) Maximum damage radius
- (3) Cumulative exposure
- (4) Social acceptability

22. Which software integrates consequence, frequency and risk visualization ?

- (1) AutoCAD
- (2) PHAST Risk
- (3) ANSYS
- (4) MATLAB

23. Toxic load exponent accounts for

- (1) Wind turbulence
- (2) Dose-response nonlinearity
- (3) Release geometry
- (4) Human immunity

24. QRA is least reliable when applied to

- (1) Established chemical plants
- (2) Novel processes
- (3) Storage terminals
- (4) Refineries



25. Acceptable risk criteria are primarily derived from

(1) Corporate policy

(2) Regulatory frameworks

(3) Public opinion

(4) Insurance requirements

26. HAZOP is fundamentally based on

(1) Statistical inference

(2) Guideword deviation logic

(3) Failure probability

(4) Accident history

27. HAZAN differs from HAZOP because it focuses on

(1) Design intent

(2) Risk ranking

(3) Qualitative deviation

(4) Operability

28. System life-cycle safety must begin at

(1) Commissioning

(2) Operation

(3) Conceptual design

(4) Maintenance

29. A safety-critical function failure can result in

(1) Reduced efficiency

(2) Catastrophic consequences

(3) Minor deviation

(4) Process delay

30. Which analysis best integrates human, hardware and software failures ?

(1) FMEA

(2) FTA

(3) System safety analysis

(4) Checklist analysis

31. SIL levels are associated with

(1) Electrical safety

(2) Instrumented protection systems

(3) Fire protection

(4) Structural design

32. In PHA, risk ranking mainly depends on

(1) Cost impact

(2) Likelihood and severity

(3) Equipment age

(4) Operator skill



33. A common limitation of FMEA is

- (1) Inability to prioritize risk
- (2) Lack of quantitative logic
- (3) Excessive focus on single failures
- (4) High data requirement

34. Safety validation ensures that

- (1) System meets operational needs
- (2) Hazards are fully eliminated
- (3) Safety requirements are met
- (4) Maintenance is simplified

35. Process safety management emphasizes

- (1) Worker discipline
- (2) Technical integrity and management systems
- (3) PPE usage
- (4) Emergency drills



36. Fail-safe design ensures that upon failure the system



- (1) Continues operation
- (2) Shuts down safely
- (3) Alerts operators
- (4) Reduces efficiency

37. Pressure relief valve sizing is governed primarily by

- (1) Normal operating pressure
- (2) Worst credible overpressure scenario
- (3) Vessel volume
- (4) Material strength

38. Plant layout safety aims to minimize

- (1) Construction cost
- (2) Domino effects
- (3) Maintenance time
- (4) Land usage



39. Electrical safety earthing mainly prevents

- (1) Equipment damage
- (2) Fire
- (3) Electric shock
- (4) Voltage fluctuation



40. Fire load is defined as

- (1) Total combustible mass
- (2) Heat release potential
- (3) Ignition temperature
- (4) Flame spread rate

41. In control system design, redundancy improves

- (1) Accuracy
- (2) Reliability
- (3) Efficiency
- (4) Speed

42. Management of Change (MoC) is critical to prevent

- (1) Equipment aging
- (2) Unintended risk introduction
- (3) Human error
- (4) Maintenance delays

43. Structural safety factor primarily addresses

- (1) Economic loss
- (2) Material uncertainty
- (3) Operational efficiency
- (4) Thermal stress

44. Fire triangle excludes

- (1) Fuel
- (2) Oxygen
- (3) Ignition source
- (4) Pressure

45. Explosion venting effectiveness depends on

- (1) Vent area and response time
- (2) Wind direction
- (3) Fuel cost
- (4) Fire brigade access

46. Intrinsically safe equipment is designed to

- (1) Prevent overheating
- (2) Limit energy release
- (3) Reduce vibration
- (4) Improve efficiency

47. Control valve failure mode selection depends on

- (1) Cost
- (2) Safety consequence
- (3) Availability
- (4) Operator preference



48. Passive fire protection differs from active because it

- (1) Requires power
- (2) Acts without activation
- (3) Needs sensors
- (4) Is maintenance-intensive

49. Design for maintainability primarily improves

- 
- (1) Capital cost
 - (2) System uptime
 - (3) Fire resistance
 - (4) Structural strength

50. The primary objective of safety interlocks is to

- (1) Improve control accuracy
 - (2) Prevent unsafe operation
 - (3) Reduce manpower
 - (4) Increase throughput
- 

51. Reliability is defined as probability of

- (1) Failure occurrence
- (2) Successful operation over time
- (3) Repair completion
- (4) System availability

52. RCM focuses on

- (1) Scheduled maintenance
- (2) Function-based maintenance
- (3) Breakdown repair
- (4) Cost reduction

53. Human error probability is influenced most by

- (1) Age
- (2) Workload and stress
- (3) Education
- (4) Gender

54. Anthropometry is essential for

- (1) Equipment reliability
 - (2) Workstation design
 - (3) Fire safety
 - (4) Risk assessment
- 

55. MSDs primarily result from

- (1) Chemical exposure
- (2) Poor posture and repetition
- (3) Noise
- (4) Radiation

56. Incident investigation aims mainly to identify

- (1) Blame
- (3) Legal liability

(2) Root causes

(4) Compensation

57. Behavioural safety focuses on

- (1) Equipment design
- (3) Process automation

(2) Worker actions

(4) Environmental control

58. Fitness-to-work assessment evaluates

- (1) Skill level
- (3) Experience

(2) Medical suitability

(4) Productivity

59. Occupational health surveillance is

- (1) Reactive
- (3) Disciplinary

(2) Preventive

(4) Optional

60. Labour welfare amenities primarily improve

- (1) Profitability
- (3) Automation

(2) Worker well-being

(4) Compliance cost

61. Hazardous waste classification depends on

- (1) Quantity only
- (3) Storage location

(2) Toxicity and reactivity

(4) Disposal cost

62. Effluent treatment plants are designed to

- (1) Neutralize all waste
- (3) Eliminate solid waste

(2) Reduce pollutants to permissible limits

(4) Recover energy

63. Life-cycle costing considers

- (1) Capital cost only
- (3) Maintenance cost only

(2) Total cost over system life

(4) Replacement cost only



64. Maintainability is measured by

(1) MTBF

(3) Failure rate

(2) MTTR

(4) Availability

65. Human reliability analysis quantifies



(1) Machine failure

(3) Training hours

(2) Operator error probability

(4) Accident cost

66. Safety performance monitoring uses

(1) Only accident data

(3) Financial reports

(2) Leading and lagging indicators

(4) Audit scores only

67. Job stress increases primarily due to

(1) High autonomy

(3) Ergonomic design

(2) Role ambiguity

(4) Adequate staffing

68. Waste minimization hierarchy prioritizes

(1) Disposal

(3) Recycling

(2) Treatment

(4) Source reduction

69. Environmental compliance failure mainly results in

(1) Productivity loss

(3) Equipment damage

(2) Legal and reputational risk

(4) Worker absenteeism

70. System availability combines

(1) Reliability and maintainability

(3) Design and operation

(2) Cost and safety

(4) Failure and repair

71. Cognitive overload primarily affects

(1) Physical endurance

(3) Vision

(2) Decision-making

(4) Hearing



72. Accident causation models emphasize

- (1) Single cause
- (2) Multiple interacting factors
- (3) Equipment failure only
- (4) Worker negligence

73. Safety management systems are aligned with

- (1) ISO 9001
- (2) ISO 45001
- (3) ISO 14001
- (4) ISO 50001

74. Risk-based maintenance prioritizes equipment based on

- (1) Age
- (2) Failure consequence and likelihood
- (3) Manufacturer recommendation
- (4) Cost only

75. Occupational diseases typically show

- (1) Immediate symptoms
- (2) Latency period
- (3) Seasonal variation
- (4) Acute exposure only

76. Waste disposal without treatment mainly affects

- (1) Air quality
- (2) Soil and groundwater
- (3) Noise levels
- (4) Lighting

77. Safety audits are most effective when conducted

- (1) After accidents
- (2) Periodically and systematically
- (3) Randomly
- (4) Only for compliance

78. Human-machine interface design reduces

- (1) Maintenance cost
- (2) Human error
- (3) Energy consumption
- (4) Fire risk

79. Environmental risk differs from safety risk because it is

- (1) Immediate
- (2) Often long-term and diffuse
- (3) Quantifiable easily
- (4) Localized

80. Ergonomic risk assessment tools include

(1) RULA and REBA

(3) HAZOP

(2) FTA and ETA

(4) QRA

81. Preventive maintenance aims to

(1) Fix failures

(3) Reduce manpower

(2) Avoid failures

(4) Increase downtime

82. Work environment interaction considers

(1) Man only

(3) Man-machine-environment

(2) Machine only

(4) Organization only

83. Noise exposure limits are expressed in

(1) Hz

(3) Pascal

(2) dB(A)

(4) Lux

84. Environmental management systems emphasize

(1) End-of-pipe solutions

(3) Legal defense

(2) Continual improvement

(4) Waste disposal

85. Incident reporting culture improves

(1) Punishment

(3) Productivity

(2) Learning and prevention

(4) Surveillance

86. A near-miss indicates :

(1) Unsafe condition without consequence

(3) Property damage

(2) Minor accident

(4) Regulatory violation

87. Musculoskeletal disorder prevention focuses on

(1) PPE only

(3) Medical treatment only

(2) Work redesign

(4) Overtime reduction



88. Emergency preparedness primarily aims to

(1) Avoid accidents

(3) Improve compliance

(2) Reduce accident consequences

(4) Increase insurance

89. Wastewater BOD indicates

(1) Toxicity

(3) Suspended solids

(2) Organic pollution load

(4) Acidity

90. Human error classification includes

(1) Slips and lapses

(3) Structural collapse

(2) Equipment failure

(4) Fire

91. Safety leadership effectiveness is reflected in

(1) Written rules

(3) Accident cost

(2) Employee safety behavior

(4) Audit frequency

92. Occupational hygiene focuses on

(1) Medical treatment

(3) Compensation

(2) Hazard anticipation and control

(4) Legal compliance

93. Safety-critical systems require

(1) Minimal redundancy

(3) Operator dependence

(2) High reliability and validation

(4) Low cost

94. Waste incineration is unsuitable for

(1) Organic waste

(3) Toxic solids

(2) High-moisture waste

(4) Medical waste

95. Risk communication is essential for

(1) Public trust

(3) Cost reduction

(2) Legal defense

(4) Insurance



96. Accident investigation reports should be

- (1) Confidential only
- (2) Blame-oriented
- (3) Fact-based and objective
- (4) Lengthy

97. Occupational exposure monitoring frequency depends on

- (1) Worker age
- (2) Risk severity
- (3) Company size
- (4) Profit margin

98. Safety training effectiveness improves when it is

- (1) Generic
- (2) Task-specific
- (3) Theoretical
- (4) Optional

99. Environmental sustainability integrates

- (1) Safety only
- (2) Economic, social, environmental aspects
- (3) Profit only
- (4) Compliance only

100. The ultimate goal of industrial safety is

- (1) Zero paperwork
- (2) Zero accidents philosophy
- (3) Regulatory compliance
- (4) Cost minimization

101. The primary reason for the evolution of industrial safety as a formal discipline during the Industrial Revolution was

- (1) Advancement in production technology
- (2) Increased capital investment
- (3) High incidence of fatal industrial accidents
- (4) Globalization of industries

102. In the hierarchy of hazard control, which measure is considered the most effective ?

- (1) Personal protective equipment
- (2) Administrative controls
- (3) Engineering controls
- (4) Hazard elimination

103. Under the Indian Factories Act, 1948, which condition makes a premise legally classified as a factory ?

- (1) Is open to the public
- (2) Employs 5 workers without using power
- (3) Use of power and employment of 10 or more workers
- (4) Produces goods for sale only

104. Which occupational disease is most commonly associated with prolonged exposure to silica dust ?

- (1) Asbestosis
- (2) Silicosis
- (3) Byssinosis
- (4) Pneumoconiosis

105. Which of the following international conventions strongly influenced occupational safety and health legislation worldwide ?

- (1) Kyoto Protocol
- (2) Basel convention
- (3) ILO Convention No. 155
- (4) Stockholm convention

106. Environmental Impact Assessment (EIA) is best described as

- (1) A post-project pollution monitoring tool
- (2) A legal penalty mechanism
- (3) A predictive tool to evaluate environmental consequences of proposed projects
- (4) An industrial waste management technique

107. Which legislation in India specifically integrates industrial safety, occupational health and environmental protection into a unified legal framework ?

- (1) Factories Act, 1948
- (2) Environment (Protection) Act, 1986
- (3) Industrial Disputes Act, 1947
- (4) Boiler's Act, 1923

108. Which of the following is primarily evaluated during consequence analysis in QRA ?

- (1) Failure probability of components
- (2) Human error probability
- (3) Impact of hazardous release on people and property
- (4) Reliability of safety systems



109. In damage modelling, thermal radiation effects are most appropriately correlated using

- (1) Overpressure-distance relationships
- (2) Probit functions
- (3) Gaussian plume equations
- (4) Event tree probabilities



110. Dense gas dispersion models are required when

- (1) The released gas is lighter than air
- (2) Atmospheric turbulence is negligible
- (3) Gas density is significantly higher than air
- (4) Wind speed is very high

111. Uncertainty in QRA results mainly arises due to

- (1) Errors in mathematical formulation only
- (2) Variability in meteorological conditions and data limitations
- (3) Software coding errors
- (4) Improper presentation of results



112. Process Hazard Analysis (PHA) is primarily conducted to

- (1) Estimate financial losses due to accidents
- (2) Identify and evaluate potential hazards in a process
- (3) Quantify individual and societal risk
- (4) Verify compliance with environmental standards

113. Which of the following techniques is considered the most qualitative hazard identification method ?



- (1) HAZAN
- (2) FTA
- (3) HAZOP
- (4) Quantitative Risk Assessment

114. In system safety engineering, the preferred order of hazard control is

- (1) Warnings → PPE → Design changes
- (2) PPE → Administrative controls → Design changes
- (3) Design elimination → Engineering controls → Administrative controls
- (4) Administrative controls → Engineering controls → PPE



115. In structural design safety, the use of partial safety factors mainly accounts for

- (1) Construction speed
- (2) Architectural requirements
- (3) Uncertainty in loads and material strength
- (4) Aesthetic considerations



116. Which of the following is the most critical design consideration for pressure vessels operating at high pressure?

- (1) External corrosion only
- (2) Hoop stress due to internal pressure
- (3) Wind load
- (4) Thermal insulation

117. In fire safety engineering, passive fire protection refers to

- 
- (1) Fire alarms and detectors
 - (2) Automatic sprinkler systems
 - (3) Fire-resistant walls and structural fireproofing
 - (4) Fire-fighting personnel

118. For a system of components connected in series, the system reliability is

- (1) Greater than the most reliable component
- (2) Equal to the arithmetic mean of component reliabilities
- (3) Product of individual component reliabilities
- (4) Sum of individual component reliabilities

119. Which configuration improves system reliability by providing redundancy?

- 
- (1) Series system
 - (2) Parallel system
 - (3) Standby-only system
 - (4) Single-path system

120. Maintainability is best defined as the

- (1) Probability that a system will not fail
- (2) Ability to predict failures accurately
- (3) Ease and speed with which a system can be restored after failure
- (4) Total maintenance cost over life

121. For a complex system with multiple subsystems, overall reliability analysis is best carried out using

(1) Reliability block diagrams and fault tree analysis

(2) Single-component failure models

(3) Visual inspection methods

(4) Simple time-based models

122. The main objective of working posture analysis is to

(1) Increase production rate

(2) Identify postures that may lead to discomfort or injury

(3) Reduce machine cycle time

(4) Improve product aesthetics

123. In workstation design, the preferred design principle is to

(1) Design for the average user

(2) Design for the smallest user

(3) Design for the tallest user

(4) Design for adjustability

124. Cognitive modelling in ergonomics mainly focuses on

(1) Muscle fatigue behaviour

(2) Biomechanical load estimation

(3) Mental processes such as perception, decision-making, and memory

(4) Physical workstation layout

125. Which measure is most effective in preventing MSDs at the workplace ?

(1) Increasing working hours

(2) Use of personal protective equipment alone

(3) Job rotation without design changes

(4) Ergonomic redesign of tasks and workstations



126. Workplace stress is most likely to increase when

- (1) Job demands exceed worker control and resources
- (2) Tasks are ergonomically designed
- (3) Adequate rest breaks are provided
- (4) Automation supports decision-making

127. Engineering controls are preferred over administrative controls because they



- (1) Depend on worker behaviour
- (2) Are cheaper to implement
- (3) Eliminate or reduce hazards at source
- (4) Require extensive training

128. Administrative controls primarily include

- (1) Machine guarding
- (2) Safety policies, procedures and training
- (3) Process redesign
- (4) Inherent safety measures

129. Fault diagnosis primarily aim to

- (1) Increase system availability
- (2) Identify the location and cause of faults
- (3) Reduce maintenance manpower
- (4) Improve system aesthetics

130. The interaction between man, machine and work environment is best addressed



through

- (1) Isolated equipment design
- (2) Post-accident training
- (3) Systems and ergonomic approach
- (4) Disciplinary actions

131. Pre-employment medical examination is mainly conducted to



- (1) Reduce employer liability
- (2) Establish baseline health status of workers
- (3) Eliminate the need for periodic health checks
- (4) Improve worker morale



132. Which of the following is a physical occupational health risk ?

- (1) Job insecurity
- (2) Shift work
- (3) Noise exposure
- (4) Work-related stress

133. Which worker group most critically requires periodic audiometric testing ?



- (1) Office staff
- (2) Laboratory analysts
- (3) Workers exposed to high noise levels
- (4) Administrative supervisors

134. The effectiveness of a health surveillance program is best evaluated by

- (1) Number of medical tests conducted
- (2) Cost of the program
- (3) Reduction in detected cases over time due to preventive action
- (4) Size of medical records maintained

135. Which of the following is an essential principle of occupational health records ?



- (1) Open public access
- (2) Confidentiality of worker health data
- (3) Short-term storage
- (4) Elimination after employment ends

136. Biological monitoring in occupational health is used to

- (1) Measure workplace air concentration only
- (2) Assess ergonomic risk
- (3) Evaluate individual exposure through biological samples
- (4) Predict accident probability

137. Long-term retention of occupational health records is important mainly for



- (1) Legal compliance and tracking chronic occupational diseases
- (2) Daily attendance management
- (3) Wage calculation
- (4) Performance appraisal

138. Which of the following is classified as a statutory welfare amenity under Labour Legislation ?

- (1) Recreational clubs
- (2) Canteen facilities
- (3) Incentive bonus
- (4) Profit sharing

139. The effectiveness of Labour Welfare amenities largely depends on

- (1) Initial capital investment only
- (2) Frequency of government inspections
- (3) Worker discipline
- (4) Proper maintenance and regular supervision

140. Which welfare amenity directly contributes to reducing fatigue and work-related stress ?

- (1) Rest rooms and shelters
- (2) Attendance registers
- (3) Time office
- (4) Disciplinary committees

141. Appointment of a welfare officer in large industrial establishments is mainly to

- (1) Enforce disciplinary rules
- (2) Represent management interests
- (3) Ensure proper implementation and upkeep of welfare measures
- (4) Conduct production planning

142. Which approach ensures long-term sustainability of Labour Welfare amenities ?

- (1) Outsourcing all welfare functions
- (2) One-time provision of facilities
- (3) Minimum statutory compliance only
- (4) Continuous evaluation and improvement of amenities

143. Which welfare amenity is most directly associated with occupational health promotion at the workplace ?

- (1) Creche facility
- (2) First-aid and medical facilities
- (3) Housing allowance
- (4) Recreational clubs

144. Industrial effluents containing high concentrations of organic matter are best treated by

- (1) Biological treatment methods
- (2) Thermal incineration
- (3) Landfilling
- (4) Physical screening



145. Which of the following is classified as a hazardous industrial waste ?

- (1) Fly ash from thermal power plants
- (2) Wastewater sludge containing heavy metals
- (3) Cooling tower blowdown
- (4) Treated domestic sewage

146. Primary treatment of industrial wastewater mainly involves

- (1) Biological oxidation
- (2) Chemical precipitation
- (3) Physical processes like screening and sedimentation
- (4) Advanced membrane separation

147. Which treatment method is most effective for removing dissolved inorganic salts from effluents ?

- (1) Activated sludge process
- (2) Reverse osmosis
- (3) Aerated lagoons
- (4) Trickling filters

148. Secure landfilling is recommended mainly for

- (1) Recyclable waste
- (2) Inert solid waste
- (3) Biodegradable municipal waste
- (4) Stabilized hazardous waste residues

149. Neutralization of acidic industrial effluents is typically achieved using

- (1) Lime or caustic soda
- (2) Activated carbon
- (3) Chlorination
- (4) Aeration

150. Which of the following is an example of physico-chemical treatment ?

- (1) Anaerobic digestion
- (2) Composting
- (3) Coagulation-flocculation
- (4) Activated sludge process



EI/FOI/616

SPACE FOR ROUGH WORK

145. Which of the following is a physical property of a substance?



146. Which of the following is a chemical property of a substance?

- (1) Ability to burn
- (2) Ability to change color
- (3) Ability to change shape
- (4) Ability to change volume

147. Which of the following is a physical property of a substance?

- (1) Ability to burn
- (2) Ability to change color
- (3) Ability to change shape
- (4) Ability to change volume

148. Which of the following is a chemical property of a substance?

- (1) Ability to burn
- (2) Ability to change color
- (3) Ability to change shape
- (4) Ability to change volume

149. Which of the following is a physical property of a substance?

- (1) Ability to burn
- (2) Ability to change color
- (3) Ability to change shape
- (4) Ability to change volume